

We're presented with a Go-Binary (PRECISE.exe) and a set of artifacts (All.zip)

```
PS C:\Users\...\Downloads> .\PRECISE.exe
-- Precise Remote Access --
This is a foolproof tool that makes sure the target is right before they get accepted by the Precise C2!
I'm not ACTUALLY going to make sure the target is right but we won't work if the target is not!
I might be the greatest offensive engineer of all time with this level of security!
At least
I think so, we will pass it and see
G000f>;0Yc$000z0%0400gB00j0g access C2 is : r0[NNs
Precise C2 will decide if you get accepted or not!
These are printed for no apparent reason! Or they might be used to d.--
Username, Expected : <#REDACTED#> got : 
Computer name, Expected : <#REDACTED#> got : MIIKIE
Date, (format YYYY-MM-DD) Expected : <#REDACTED#> got : 2026-07-21
Volume serial, (format XXXX-XXXX) Expected : <#REDACTED#> got : 
Keyboard layout, (Windows KLID) Expected : <#REDACTED#> got : 00000409
Connection failed, what a surprise man, welp at least the secret was sent somewhere %v dial tcp 10.10.10.10:4444: connec
tex: A connection attempt failed because the connected party did not properly respond after a period of time, or establi
shed connection failed because connected host has failed to respond.
Panic: runtime error: invalid memory address or nil pointer dereference
To be precise is to be patient...
.
.
..
...
PS C:\Users\...\Downloads>
```

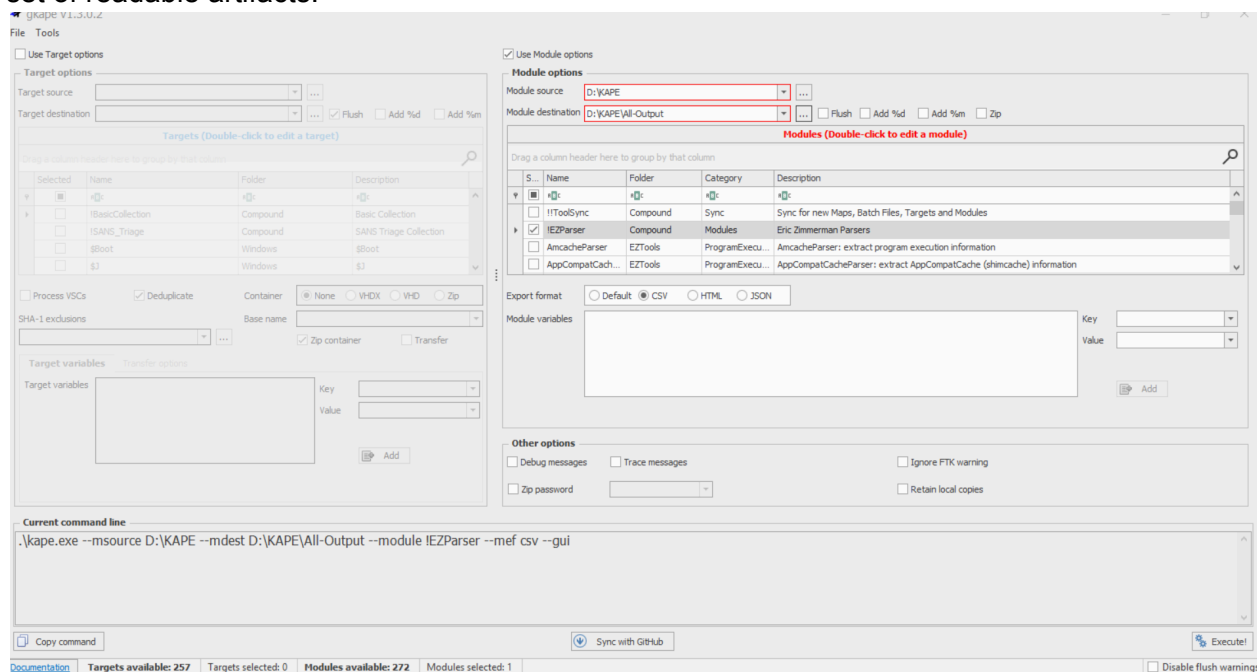
From this we can determine that this tool is used to ensure that a target connecting to C2 is 'legitimate', we receive garbled text that only resolves into the real key when the target details are correct.

The program prints out five values: Username, Computer Name, Date, Serial Volume ID, Keyboard Layout ID. The program is clearly expecting for these values to be correctly set before it allows the target device to "connect" to the C2.

Within all.zip (the artifacts zipfile), various yet limited artifacts are provided, including all registry hives (System/User/Other), LNK + Jumplists, EventLogs, ETL Traces. Out of these, Only the Hives (System/User) and .lnk files and jumplists are needed, the artifacts provided are intentionally heavy and a player is expected to sift through them all to find the respective values.

For finding username, machine name, date: Players are expected to make use of KAPE to use the **!EZParser** module or individually use EZTools (LECmd, RECmd, Registry Explorer):

Firstly, using KAPE to easily parse all artifacts with the **!EZParser** module. This results in a clean set of readable artifacts.



	EventLogs	7/21/2026 12:23 AM	File folder	
	FileDeletion	7/21/2026 12:24 AM	File folder	
	FileFolderAccess	7/21/2026 12:25 AM	File folder	
	ProgramExecution	7/21/2026 12:23 AM	File folder	
	Registry	7/21/2026 12:25 AM	File folder	
	SQLDatabases	7/21/2026 12:25 AM	File folder	
	SRUMDatabase	7/21/2026 12:25 AM	File folder	
	SUMDatabase	7/21/2026 12:25 AM	File folder	
	2026-07-20T18_53_53_3581471_Conso...	7/21/2026 12:25 AM	Text Document	8 KB

For username: Hivepath already displays `C\Users\driver\*`, if not explicit enough, it could also be found at the `RegisteredOwner` ValueName where ValueData is `driver` within RECmnds's expected parsed output.

20260720185441	7/21/2026 12:25 AM	File folder	
20260720185441_RECcmd_Batch_Kroll_...	7/21/2026 12:25 AM	Comma Separated...	1,220 KB
639	t),System Info,ROOT\Microsoft\Windows NT\CurrentVersion,RegisteredOwner,RegSz,driver,		

For machine name: `ComputerName` ValueName stores ValueData `DESKTOP-4MJ3RFH` within RECmnds's parsed output.

2107	nfo,ROOT\ControlSet001\Control\DeviceClasses,Multiple,(plugin),Type: STORAGE Name: Volume SerialNumber: {003b44eb-831f-11f1-9dc1-806e6f6
2108	nfo,ROOT\ControlSet001\Control\DeviceClasses,Multiple,(plugin),Type: STORAGE Name: Volume SerialNumber: {003b44eb-831f-11f1-9dc1-806e6f6
2109	nfo,ROOT\ControlSet001\Control\DeviceClasses,Multiple,(plugin),Type: USB Name: VID_0627&PID_0001 SerialNumber: 28754-0000:00:1d.7-1,Time
2110	ystem Info,ROOT\ControlSet001\Control\ComputerName\ComputerName,ComputerName,RegSz,DESKTOP-4MJ3RFH,,Name of computer used by the user,F
2111	Info,ROOT\ControlSet001\Control\Class\{4d36e972-e325-11ce-bfc1-08002be10318},Multiple,(plugin),DriveName: Realtek RTL8139C+ Fast Ethern
2112	Info,ROOT\ControlSet001\Control\Class\{4d36e972-e325-11ce-bfc1-08002be10318},Multiple,(plugin),DriveName: Microsoft Kernel Debug Networ
2113	Info,ROOT\ControlSet001\Control\Class\{4d36e972-e325-11ce-bfc1-08002be10318},Multiple,(plugin),DriveName: WAN Miniport (SSTP) DriverDat

For Date: `InstallTime` & `InstallDate` ValueName stores the same ValueData `2026-07-19` within RECmnds's parsed output.

641	m Info (Current),System Info,ROOT\Microsoft\Windows NT\CurrentVersion,InstallTime,RegQword,2026-07-19 02:09:59.6611327,,
642	m Info (Current),System Info,ROOT\Microsoft\Windows NT\CurrentVersion,ProductName,RegSz,Windows 10 Home,,Current OS nam
643	m Info (Current),System Info,ROOT\Microsoft\Windows NT\CurrentVersion,InstallDate,RegDword,2026-07-19 02:09:59.0000000,,

For KLID: Players are expected to use Eric Zimmerman's Registry Explorer and load the NTUSER.dat hive directly. The expected value is 00020409 (US-International), but it doesn't live in Preload. Preload\1 instead shows d0010809 the d001 prefix marks this as a substitute placeholder, not a real keyboard ID. The real KLID is resolved in Keyboard Layout\Substitutes, where the value named d0010809 points to 00020409. The second entry, Preload\2 = 00000809, is the secondary layout (UK English).

Preload

Key name	# values	# subkeys	Last write time
D:\KAPE\All\All\C\Users\driver\NTUSER...	=	=	=
ROOT	0	11	2026-07-19 13:3
Keyboard Layout	0	3	2026-07-19 02:1
Preload	2	0	2026-07-19 02:1
D:\KAPE\All\All\C\Users\driver\NTUSER...	0	11	2026-07-19 13:3
ROOT	0	3	2026-07-19 02:1
Keyboard Layout	0	3	2026-07-19 02:1
Preload	2	0	2026-07-19 02:1

Drag a column header here to group by that column

Value Name	Value Type	Data
2	RegSz	00000809
1	RegSz	d0010809

Registry hives (2) Available bookmarks (62/0)

Substitutes

Key name	# values	# subkeys	Last write time
D:\KAPE\All\All\C\Users\driver\NTUSER...	=	=	=
ROOT	0	11	2026-07-19 13:3
Keyboard Layout	0	3	2026-07-19 02:1
Substitutes	1	0	2026-07-19 02:1
D:\KAPE\All\All\C\Users\driver\NTUSER...	0	11	2026-07-19 13:3
ROOT	0	3	2026-07-19 02:1
Keyboard Layout	0	3	2026-07-19 02:1
Substitutes	1	0	2026-07-19 02:1

Values

Drag a column header here to group by that column

Value Name	Value Type	Data
d0010809	RegSz	00020409

Keyboard Layout	0	3
Preload	2	0
Substitutes	1	0

Side note: When loading the NTUSER.dat hive into RegExplorer, It'll be flagged as a dirty hive, it is up to the player to replay the logs or proceed with loading the dirty hive anyways; This happens because it was collected while pending changes still sat in the transaction logs (.LOG1/.LOG2) unwritten, which is normal for a live KAPE collection. The intended values (Keyboard Layout\Substitutes) and (Keyboard Layout\Preload) exist properly and load either way the player decides to load the hive.

For Serial Volume ID: `VolumeSerialNumber` shows: 7A50D654 -> 7A50-D654 when LECmd parses given artifacts into a csv.

	ISBECmd_Messages.txt	7/21/2026 12:25 AM	Text Document	6 KB
	20260720185433_AutomaticDestinati...	7/21/2026 12:24 AM	Comma Separated...	14 KB
	20260720185433_CustomDestinations....	7/21/2026 12:24 AM	Comma Separated...	8 KB
☒ 	20260720185436_LECmd_Output.csv	7/21/2026 12:24 AM	Comma Separated...	8 KB
	driver_NTUSER.csv	7/21/2026 12:25 AM	Comma Separated...	1 KB
	driver_UsrClass.csv	7/21/2026 12:25 AM	Comma Separated...	10 KB

```
e),7A50D654,,C:\Users\driver\Desktop\All,,,  
xed storage media (Hard drive),7A50D654,,C:\  
dia (Hard drive),7A50D654,,C:\Users\driver\
```

Now since we have all the required values needed to impersonate the target system:

Username: driver

Computer_Name: DESKTOP-4MJ3RFH

Date: 2026-07-19

KLID: 00020409

Serial Volume ID: 7A50-D654,

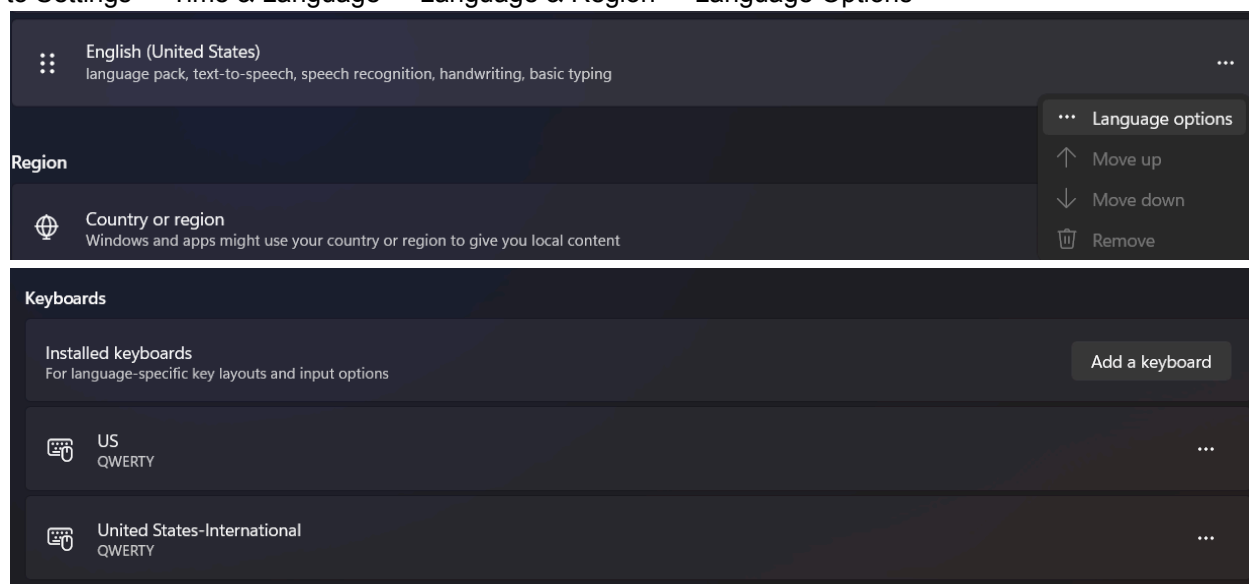
First, a quick reboot after using [VolumeID](#) to change our Serial Volume ID, you require a reboot for the Serial volume ID cache and a restart refreshes it, loading the newly set volume ID.

```
PS C:\Users\... \Downloads> .\volumeid.exe C: 7A50-D654  
  
VolumeId v2.1 - Set disk volume id  
Copyright (C) 1997-2016 Mark Russinovich  
Sysinternals - www.sysinternals.com  
  
Volume ID for drive C: updated to 7a50-d654  
PS C:\Users\... \Downloads>
```

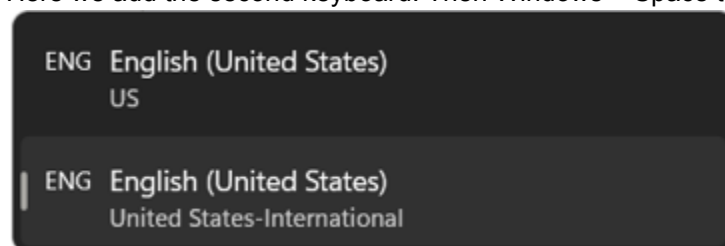
Checking if Serial Volume ID actually changed...:

```
PS C:\Users\... \Downloads> cmd /c vol C:  
Volume in drive C is Windows  
Volume Serial Number is 7A50-D654  
PS C:\Users\... \Downloads>
```

For the KLID; Since we know 00020409 Represents United States (International), we can add it by going to Settings -> Time & Language -> Language & Region -> Language Options



Here we add the second keyboard. Then Windows + Space to actually swap to it.



For the rest, we can simply just change session env vars:

Side note: this method of using Set-Date WILL silently fail Automatic time sync is on, it is expected to be off or the date can also be changed via system settings. Same effect.

```
PS C:\WINDOWS\system32> cd ../../
PS C:\> cd .\Users\_____.\Downloads\
PS C:\Users\_____.\Downloads> Set-Date "2026-07-19"
>> $env:USERNAME = "driver"
>> $env:COMPUTERNAME = "DESKTOP-4MJ3RFH"

Sunday, July 19, 2026 12:00:00 AM
```

By running PRECISE.exe now

```
PS C:\Users\...\Downloads> .\PRECISE.exe
-- Precise Remote Access --
This is a foolproof tool that makes sure the target is right before they get accepted by the Precise C2!
I'm not ACTUALLY going to make sure the target is right but we won't work if the target is not!
I might be the greatest offensive engineer of all time with this level of security!
At least
I think so, we will pass it and see
Vf^000r0000m,00000m0(0Hm0to access C2 is : u0.o^0[
Precise C2 will decide if you get accepted or not!
These are printed for no apparent reason! Or they might be used to d.--
Username, Expected : <#REDACTED#> got : <#REDACTED#>
Computer name, Expected : <#REDACTED#> got : <#REDACTED#>
Date, (format YYYY-MM-DD) Expected : <#REDACTED#> got : 2026-07-21
Volume serial, (format XXXX-XXXX) Expected : <#REDACTED#> got : <#REDACTED#>
Keyboard layout, (Windows KLID) Expected : <#REDACTED#> got : <#REDACTED#>
Connection failed, what a surprise man, welp at least the secret was sent somewhere %v dial tcp 10.10.10.10:4444: connec
tex: A connection attempt failed because the connected party did not properly respond after a period of time, or establi
shed connection failed because connected host has failed to respond.
Panic: runtime error: invalid memory address or nil pointer dereference
To be precise is to be patient...
.
..
...
```

We can see that the program expecting <Redacted> Values actually got <Redacted Values>, Indicating that those are the correct values.

For the date, because that one value doesn't match, the derived key is wrong and the ciphertext stays garbled. the five values aren't checked independently, they're all fed into the key derivation, so the flag only decrypts when all five match simultaneously.

```
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I'm not ACTUALLY going to make sure the target is right but we won't work if the target is not!
I might be the greatest offensive engineer of all time with this level of security!
At least
I think so, we will pass it and see
Software ran, secret code to access C2 is : L3AK{
Precise C2 will decide if you get accepted or not!
These are printed for no apparent reason! Or they might be used to d.--
Username, Expected : <#REDACTED#> got : <#REDACTED#>
Computer name, Expected : <#REDACTED#> got : <#REDACTED#>
Date, (format YYYY-MM-DD) Expected : <#REDACTED#> got : <#REDACTED#>
Volume serial, (format XXXX-XXXX) Expected : <#REDACTED#> got : <#REDACTED#>
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